

Technical Reading Notes

SABR Models for Synthetic FX Rates

Initial focused study (2-day reading), with main emphasis on Appendix A

Master 2 El Karoui — Working notes

April 14, 2026

Contents

1	Title and introduction	2
2	Global roadmap of the paper	2
3	Core model setup	3
3.1	Forward FX rates and measure choice	3
4	Cross FX construction and key intuition	3
4.1	Definition and structural identity	3
4.2	Itô-based dynamics	4
4.3	Main intuition behind (4)	4
5	Main result of the paper	4
6	Main focus — Appendix A	5
6.1	6.1 Starting point: the general multifactor stochastic volatility model	5
6.2	6.2 Why one wants a reduced effective description	5
6.3	6.3 Definition and role of joint density and marginal density	5
6.4	6.4 The conservation law	5
6.5	6.5 The moments $Q^{(k)}$ and why they are introduced	6
6.6	6.6 The asymptotic expansion in ε	6
6.7	6.7 The change of variables and its role	6
6.8	6.8 The constitutive relation linking $Q^{(2)}$ and $Q^{(0)}$	6
6.9	6.9 The effective forward equation	7
6.10	6.10 Why this effective forward equation corresponds to a SABR model	7
6.11	6.11 Final effective SABR parameters and interpretation	7
7	What I understand well at this stage	7
8	What remains technically difficult	8
9	Conclusion	8

1 Title and introduction

This paper studies how to infer a *cross-FX volatility smile* from two more liquid FX option markets, in a SABR framework. The object is the cross forward rate

$$\tilde{X}_{12} = \frac{\tilde{X}_{10}}{\tilde{X}_{20}},$$

where currency 0 is a base currency (typically USD).

Why this matters in practice:

- Cross FX options are economically important for corporates and macro investors (hedging and relative-value positioning between non-USD currencies).
- Liquidity is often concentrated in “currency/base” pairs (e.g. EURUSD, GBPUSD), while direct cross pairs may be less liquid.
- If we can reconstruct a reliable cross smile from liquid pairs, we can quote, hedge, and risk-manage cross options with better consistency.

The paper’s key message is that, under lognormal SABR ($\beta = 1$) assumptions, the cross FX can be represented by an *effective* lognormal SABR model, accurate through $O(\varepsilon^2)$, i.e. the same asymptotic order as classical SABR implied-volatility formulas.

2 Global roadmap of the paper

Main text

The main text sets up the financial problem: from two FX rates against a common base currency, derive the smile dynamics of the cross rate. It shows (using Itô and measure choice) that the cross dynamics are driven by a linear combination of the two component stochastic-vol drivers. It then states effective SABR parameters and discusses practical correlation structures (single stochastic vol, near-independent volatility limit, and multi-currency consistency ideas).

Appendix A

Appendix A is the mathematical engine. It analyzes a general multi-factor stochastic-vol model:

$$d\tilde{X} = \varepsilon \tilde{X} \sum_{j=1}^n \lambda_j \tilde{A}_j dW_j, \quad d\tilde{A}_j = \varepsilon \nu_j \tilde{A}_j dZ_j,$$

with full correlation structure. Through singular perturbation and moment-closure style arguments, it derives an effective one-dimensional forward equation for the marginal density of $\tilde{X}$, then matches this equation to a lognormal SABR forward equation to read off effective SABR parameters.

Appendix B

Appendix B specializes Appendix A to the cross-FX case $n = 2$, with $\lambda_1, \lambda_2 \in \{\pm 1\}$ (ratio/product/reverse pairs), and gives explicit effective parameters in general and simplified correlation regimes.

Global mathematical strategy

At a high level:

1. Start from high-dimensional stochastic-vol dynamics;
2. Derive an equation for the *marginal* law of the tradable $\tilde{X}$;
3. Build an asymptotic constitutive relation $Q^{(2)} \leftrightarrow Q^{(0)}$;
4. Close the marginal equation and identify it with an effective SABR forward PDE.

This is a reduction principle: replace a complicated model by an effective one preserving European prices to second order.

3 Core model setup

3.1 Forward FX rates and measure choice

For a fixed settlement date T_{set} , define forward FX rates $\tilde{X}_{i0}$ (currency i in terms of base currency 0). Under the forward measure associated with the denominating currency numeraire, forward FX rates are martingales, so drift terms vanish.

For $i = 1, 2$, the paper uses:

$$d\tilde{X}_{i0} = \varepsilon \tilde{A}_i \tilde{X}_{i0} dW_i, \quad (1)$$

$$d\tilde{A}_i = \varepsilon \nu_i \tilde{A}_i dZ_i, \quad (2)$$

$$dW_i dZ_i = \rho_{ii} dt. \quad (3)$$

Interpretation term by term.

- $\tilde{X}_{i0}$: forward FX level to be priced/hedged.
- $\tilde{A}_i$: instantaneous stochastic volatility factor.
- ν_i : vol-of-vol intensity (how fast volatility itself fluctuates).
- ρ_{ii} : spot-vol leverage correlation controlling skew sign and magnitude.
- ε : asymptotic scaling parameter (bookkeeping for small-vol-of-vol expansion).

Modeling intuition. Equation (1) says return noise is proportional to current level (lognormal setting, $\beta = 1$). Equation (2) says volatility is itself random and multiplicative. Their correlation ρ_{ii} transmits volatility shocks into return asymmetry, generating smile/skew.

4 Cross FX construction and key intuition

4.1 Definition and structural identity

The cross forward rate between currencies 1 and 2 is

$$\tilde{X}_{12} = \frac{\tilde{X}_{10}}{\tilde{X}_{20}}.$$

Financially, one unit of currency 1 in currency 2 can be synthesized by buying currency 1 vs base and selling currency 2 vs base; this naturally gives a ratio.

4.2 Itô-based dynamics

Applying Itô:

$$d\tilde{X}_{12} = \frac{d\tilde{X}_{10}}{\tilde{X}_{20}} - \frac{\tilde{X}_{10}}{\tilde{X}_{20}^2} d\tilde{X}_{20} + \text{drift terms}.$$

In the proper forward measure for currency 2 settlement, $\tilde{X}_{12}$ is a martingale, so drift disappears:

$$d\tilde{X}_{12} = \varepsilon \tilde{X}_{12} \left(\tilde{A}_1 dW_1 - \tilde{A}_2 dW_2 \right). \quad (4)$$

4.3 Main intuition behind (4)

- **Difference of drivers.** The cross is a ratio, so its log-return is approximately the difference of component log-returns. This is why drivers enter with opposite signs.
- **Role of correlation** $c = dW_1 dW_2 / dt$. If dW_1 and dW_2 move together (c high), part of the noise cancels in the difference; effective cross variance is reduced.
- **Effective volatility idea.** Instantaneous cross variance is driven by

$$\Delta^2(\mathbf{A}) = \tilde{A}_1^2 - 2c \tilde{A}_1 \tilde{A}_2 + \tilde{A}_2^2,$$

which already looks like a quadratic “effective volatility norm”. This is the seed of the later effective SABR reduction.

5 Main result of the paper

The central practical conclusion is:

For each expiry T_{ex} , the cross FX distribution (and therefore European option prices) is matched, up to $O(\varepsilon^2)$, by a lognormal SABR model with effective parameters $(\alpha_{\text{eff}}, \rho_{\text{eff}}, \nu_{\text{eff}})$.

In the paper’s notation:

$$d\tilde{X} = \varepsilon \tilde{A} \tilde{X} dW, \quad (5)$$

$$d\tilde{A} = \varepsilon \nu_{\text{eff}} \tilde{A} dZ, \quad dW dZ = \rho_{\text{eff}} dt, \quad (6)$$

with

$$\alpha_{\text{eff}} = \tilde{A}(0) = \Delta(\mathbf{a}) \exp\left(\frac{1}{4} \varepsilon^2 \Gamma(\mathbf{a}) T_{\text{ex}}\right), \quad \rho_{\text{eff}} = \frac{\Lambda(\mathbf{a})}{\sqrt{\Sigma(\mathbf{a})}}, \quad \nu_{\text{eff}} = \sqrt{\Sigma(\mathbf{a})},$$

where $\mathbf{a}$ depends on the component SABR parameters and cross structure (λ_j signs).

Why useful in practice. Instead of simulating/calibrating a high-dimensional stochastic-vol system every time, one can:

1. calibrate liquid pairs,
2. compute effective cross SABR parameters analytically,
3. price and hedge cross options with standard SABR tools.

This gives consistency plus computational tractability.

6 Main focus — Appendix A

6.1 6.1 Starting point: the general multifactor stochastic volatility model

Appendix A starts from

$$d\tilde{X} = \varepsilon \tilde{X} \sum_{j=1}^n \lambda_j \tilde{A}_j dW_j, \quad (7)$$

$$d\tilde{A}_j = \varepsilon \nu_j \tilde{A}_j dZ_j, \quad (8)$$

with full covariance specification for W_i, Z_j .

Conceptual idea: do not hard-code FX specifics too early. Build a general reduction theorem that can later be specialized to cross FX ($n = 2, \lambda_1 = 1, \lambda_2 = -1$).

Technical manipulation: introduce a small parameter ε so one can systematically sort terms by asymptotic order.

6.2 6.2 Why one wants a reduced effective description

The full state variable is $(\tilde{X}, \tilde{A}_1, \dots, \tilde{A}_n)$, so the exact PDE is $n + 1$ -dimensional.

Problem: European options depend only on $\tilde{X}(T)$, not on terminal $\tilde{A}$ -components individually.

Goal: obtain a closed equation for the marginal law of $\tilde{X}$ only.

Intuition: this is like homogenization/effective medium thinking: keep the pricing-relevant projection and absorb latent-factor impact into corrected coefficients.

6.3 6.3 Definition and role of joint density and marginal density

The appendix defines the joint transition density $p(t, x, \boldsymbol{\alpha}; T, X, \mathbf{A})$, then moments

$$Q^{(k)}(t, x, \boldsymbol{\alpha}; T, X) = \int \Delta^k(\mathbf{A}) p(\cdot) d\mathbf{A}.$$

In particular,

$$Q^{(0)} = \int p d\mathbf{A}$$

is the marginal density of $\tilde{X}(T)$.

Why this helps:

- $Q^{(0)}$ directly prices vanilla options;
- higher moments like $Q^{(2)}$ encode how latent volatility factors feed into diffusion strength.

6.4 6.4 The conservation law

Integrating the forward Kolmogorov equation of p over $\mathbf{A}$, derivative terms in A_j vanish by boundary behavior (perfect derivatives), yielding:

$$\partial_T Q^{(0)} = \frac{1}{2} \varepsilon^2 \partial_{XX} (X^2 Q^{(2)}).$$

Main conceptual point: this is a continuity/conservation equation for probability mass in X -space.

What is missing: closure. The PDE for $Q^{(0)}$ involves $Q^{(2)}$, so we need a constitutive relation $Q^{(2)} = F(Q^{(0)})$.

6.5 6.5 The moments $Q^{(k)}$ and why they are introduced

The moment family $Q^{(k)}$ is introduced precisely to organize closure.

Technical role: $Q^{(2)}$ corresponds to the local variance weight Δ^2 , which multiplies the second derivative in the forward equation.

Intuition: if Δ were deterministic, $Q^{(2)} = \Delta^2 Q^{(0)}$ exactly. With stochastic Δ , one expects corrections around this structure; Appendix A computes these corrections to second order.

6.6 6.6 The asymptotic expansion in ε

The backward equation for $Q^{(k)}$ is transformed and expanded order by order in ε . The key approximation statement is:

$$Q^{(2)} = \Delta^2(\alpha) e^{\varepsilon^2(-\Gamma+\Sigma)\tau} \left(1 + 2\varepsilon z + \varepsilon^2 \Sigma z^2\right) Q^{(0)} \quad \text{through } O(\varepsilon^2),$$

with $\tau = T - t$, $z = \frac{\log(X/x)}{\varepsilon \Delta(\alpha)}$, and explicit coefficient functionals Γ, Σ .

Conceptual reading: the random-vol system induces:

- a level renormalization (exponential factor),
- skew-type first-order correction ($2\varepsilon z$),
- curvature/vol-of-vol second-order correction ($\varepsilon^2 z^2$ -term).

6.7 6.7 The change of variables and its role

The normalized log-moneyness

$$z = \frac{\log(X/x)}{\varepsilon \Delta(\alpha)}$$

is the central variable change.

Why introduced:

- rescales strike displacement to $O(1)$ in the perturbative regime,
- isolates the main diffusion scale Δ ,
- makes the transformed equations for $k = 0$ and $k = 2$ structurally comparable.

Technical vs conceptual distinction:

- Technical: derivative transformations are algebraically heavy.
- Conceptual: this scaling is what reveals a near-universal corrected diffusion form.

6.8 6.8 The constitutive relation linking $Q^{(2)}$ and $Q^{(0)}$

Appendix A then performs two auxiliary transformations ($S^{(k)}$, then U , then H) to map the $k = 0$ equation into the $k = 2$ equation up to $O(\varepsilon^2)$, which yields the closure relation above.

Main idea: prove *equivalence of operators* after suitable gauge-like transforms.

Why this is powerful: once $Q^{(2)}$ is expressed in terms of $Q^{(0)}$, the marginal pricing PDE closes and dimension collapses from $n + 1$ to 1.

6.9 6.9 The effective forward equation

Substituting the constitutive relation into the conservation law gives:

$$\partial_T Q^{(0)} = \frac{1}{2} \varepsilon^2 \Delta^2 e^{\varepsilon^2 (\Gamma + \Upsilon) T} \partial_{XX} \left[\left(1 + 2\varepsilon \Lambda z + \varepsilon^2 \Sigma z^2 \right) X^2 Q^{(0)} \right],$$

with the paper's explicit coefficient definitions ($\Delta, \Lambda, \Sigma, \Gamma$, via g, ϕ, κ combinations).

Intuition: this is a one-factor diffusion in X , but with carefully corrected local variance structure carrying the memory of hidden vol factors.

6.10 6.10 Why this effective forward equation corresponds to a SABR model

The appendix compares this PDE with the known asymptotic forward PDE for lognormal SABR:

$$\partial_T Q_{\text{SABR}} \sim \frac{1}{2} \varepsilon^2 \alpha^2 e^{\varepsilon^2 \rho \nu T} \partial_{XX} \left[\left(1 + 2\varepsilon \rho \nu z + \varepsilon^2 \nu^2 z^2 \right) X^2 Q_{\text{SABR}} \right].$$

Matching coefficients term by term yields an *effective SABR triplet*.

Conceptual point: the reduced model is not guessed; it is identified by PDE coefficient matching at the same asymptotic accuracy order.

6.11 6.11 Final effective SABR parameters and interpretation

For a fixed expiry T_{ex} :

$$\alpha_{\text{eff}} = \Delta e^{\frac{1}{4} \varepsilon^2 \Gamma T_{\text{ex}}}, \quad \rho_{\text{eff}} = \frac{\Lambda}{\sqrt{\Sigma}}, \quad \nu_{\text{eff}} = \sqrt{\Sigma}.$$

Interpretation:

- Δ : effective instantaneous volatility level from the covariance-weighted combination of component vols;
- ρ_{eff} : effective leverage, combining spot-vol correlations and loading asymmetry;
- ν_{eff} : effective vol-of-vol of the synthetic cross.

Γ shifts the effective level via an expiry-dependent renormalization.

Big-picture takeaway on Appendix A: it is a principled asymptotic model-reduction pipeline:

multi-factor stochastic vol $\Rightarrow$ closed marginal forward PDE $\Rightarrow$ effective SABR parameters.

This is the main conceptual contribution I currently retain from the technical core.

7 What I understand well at this stage

After two days of work, I can state with confidence:

- the paper's objective and market motivation (synthetic cross smile from liquid pairs);
- the Itô/measure logic producing cross dynamics as a signed combination of component drivers;
- the effective-SABR reduction logic: closure of marginal density equation plus PDE matching;
- Appendix A's role as the derivation backbone, not just a technical appendix.

8 What remains technically difficult

Still under active study:

- some detailed manipulations in the $S^{(k)} \rightarrow U \rightarrow H$ transformation chain (bookkeeping of neglected terms and order tracking);
- full intuition for each coefficient contribution in κ , especially mixed-correlation terms;
- deeper practical calibration implications across expiries (because α_{eff} is expiry-dependent in the mapped representation);
- selected parts of Appendix B and implementation details for robust desk calibration workflows.

9 Conclusion

The paper provides an elegant and practical bridge between liquid FX smiles and less liquid cross-FX smiles. Its key strength is combining financial consistency (cross construction and numeraire logic) with asymptotic rigor (effective equation and SABR identification). At this stage, I have a clear global understanding and a genuinely focused technical grasp of Appendix A, while some coefficient-level derivations and calibration subtleties remain work in progress.